

OSINT Reconnaissance & Enumeration Report

Target Domain: example.com

Assessor Environment: Kali Linux (VirtualBox VM)

Date: September 9, 2026

1. Executive Summary

This technical report documents the results of an active and passive DNS reconnaissance exercise conducted against example.com. The main objective was to map the target's external digital footprint, identify exposed subdomains, evaluate DNS record hygiene, and uncover public information disclosures without impacting production services.

2. Lab Setup & Tools Environment

Environment Configuration:

- **Target Domain (Victim):** example.com
- **Attacker Machine:** Kali Linux running inside a VirtualBox Virtual Machine (VM).

Tools Utilized:

- **theHarvester:** Leveraged for passive Open Source Intelligence (OSINT). It gathers email addresses, employee names, public subdomains, and Autonomous System Numbers (ASNs) from search engines, and threat intelligence sources without sending direct traffic to the target infrastructure.
- **Dnsrecon:** Utilized for active DNS record analysis. It performs direct queries against authoritative name servers to extract A, AAAA, MX, NS, SOA, and TXT records,

3. Reconnaissance Methodology

1. Phase 1: Passive Reconnaissance (OSINT)

Executed theHarvester to query multiple public data sources. This phase focused on mapping the external border and discovering publicly indexed endpoints and personnel emails.

2. Phase 2: Active DNS Enumeration

Executed Dnsrecon directly against the target's DNS servers. Extracted comprehensive record types, identified mail exchange hosts, and verified whether zone transfer misconfigurations were present.

4. Summary of Key Findings

Asset / Subdomain	Record Type	Discovery Tool	Details & Risk Context
mail.example.com	MX / A Record	Dnsrecon	Primary mail gateway (IP resolved)
hera.ns.cloudflare.com	NS/ A Record	Dnsrecon	Authoritative DNS server
dev.example.com	A Record	theHarvester	Indexed staging server (Publicly accessible)
admin.example.com	A Record	Dnsrecon	Exposed management login portal
example.com	TXT Record	Dnsrecon	reveal active use of third-party
admin-2@example.com	Email Address	theHarvester	Publicly leaked admin account

5. Threat Analysis & Risk Evaluation

5.1 Exposed Management & Staging Endpoints

Risk Level: HIGH RISK

Subdomains such as dev.example.com and admin.example.com were identified. Staging environments frequently harbor unpatched code or weaker authentication controls, making them primary targets for initial access.

5.2 Employee Email Disclosures

Risk Level: MEDIUM RISK

Corporate emails discovered via theHarvester provide attackers with actionable targets for Spear-Phishing and Social Engineering campaigns.

6. Remediation & Hardening Recommendations

1. **Restrict Access to Administrative Interfaces:** Place internal administration panels (e.g., admin.example.com, dev.example.com) behind an authenticated VPN or restrict access via IP whitelisting.
2. **Disable Unnecessary Zone Transfers:** Ensure authoritative Name Servers are explicitly configured to decline unauthorized AXFR requests.
3. **Implement Mail Protection Policies:** Enforce strict SPF, DKIM, and DMARC policies in TXT records to protect corporate domains from email spoofing.